EXPERIMENT 02

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clc;
clear;
close all;

%% MICROWAVE TRANSMISSION LINES AND TWO-PORT NETWORKS
% S-Parameter Analysis

c = 3e8;                % Speed of light (m/s)
er = 2.1;               % Relative permittivity
Zref = 50;              % Reference impedance (ohms)

% Coaxial transmission line dimensions
a_inner = 0.5e-3;       % Inner conductor radius (m)
b_outer = 1.65e-3;      % Outer conductor inner radius (m)
L = 0.2;                % Transmission line length (m)

%% OBJECTIVE 1: MODEL THE COAXIAL TRANSMISSION LINE

x = linspace(-b_outer,b_outer,100);
y = linspace(-b_outer,b_outer,100);
[X,Y] = meshgrid(x,y);

R = sqrt(X.^2 + Y.^2);

% Electric field for an assumed 1 V potential difference
E = 1 ./ (R .* log(b_outer/a_inner));

% Field exists only in the dielectric region
E(R < a_inner | R > b_outer) = NaN;

%% OBJECTIVE 2: CHARACTERISTIC IMPEDANCE AND PROPAGATION CONSTANT
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% Approximate characteristic impedance of a coaxial line
Z0 = (60/sqrt(er))*log(b_outer/a_inner);

% Frequency range: 1 GHz to 10 GHz
f = linspace(1e9,10e9,600);
w = 2*pi*f;

% Phase and attenuation constants
beta = w*sqrt(er)/c;              % rad/m

% Illustrative conductor and dielectric losses
alpha_c = 0.035*sqrt(f/1e9);      % Np/m
tan_delta = 0.001;
alpha_d = beta*tan_delta/2;       % Np/m

alpha = alpha_c + alpha_d;
gamma = alpha + 1i*beta;

fprintf('TRANSMISSION LINE PARAMETERS\n');
fprintf('----------------------------\n');
fprintf('Characteristic impedance = %.3f ohms\n',Z0);
fprintf('Line length = %.3f m\n',L);
fprintf('Frequency range = 1 to 10 GHz\n');
fprintf('Reference impedance = %.1f ohms\n',Zref);

%% OBJECTIVE 3: CALCULATE S-PARAMETERS

S11 = zeros(size(f));
S21 = zeros(size(f));
S12 = zeros(size(f));
S22 = zeros(size(f));
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for k = 1:length(f)

    % ABCD parameters of the transmission line
    A = cosh(gamma(k)*L);
    B = Z0*sinh(gamma(k)*L);
    C = sinh(gamma(k)*L)/Z0;
    D = cosh(gamma(k)*L);

    den = A + B/Zref + C*Zref + D;

    % Scattering parameters
    S11(k) = (A + B/Zref - C*Zref - D)/den;
    S21(k) = 2/den;
    S12(k) = 2*(A*D - B*C)/den;
    S22(k) = (-A + B/Zref - C*Zref + D)/den;

end

%% OBJECTIVE 4: RETURN LOSS, INSERTION LOSS AND EFFICIENCY

S11_dB = 20*log10(max(abs(S11),1e-12));
S21_dB = 20*log10(max(abs(S21),1e-12));
S12_dB = 20*log10(max(abs(S12),1e-12));
S22_dB = 20*log10(max(abs(S22),1e-12));

ReturnLoss = -S11_dB;
InsertionLoss = -S21_dB;

Efficiency = abs(S21).^2*100;

fprintf('\nS-PARAMETER RESULTS AT 5 GHz\n');
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fprintf('----------------------------\n');

[~,k5] = min(abs(f-5e9));

fprintf('|S11| = %.4f\n',abs(S11(k5)));
fprintf('|S21| = %.4f\n',abs(S21(k5)));
fprintf('|S12| = %.4f\n',abs(S12(k5)));
fprintf('|S22| = %.4f\n',abs(S22(k5)));

fprintf('Return loss = %.3f dB\n',ReturnLoss(k5));
fprintf('Insertion loss = %.3f dB\n',InsertionLoss(k5));
fprintf('Transmission efficiency = %.2f %%\n',Efficiency(k5));

%% CREATE ONE FIGURE WITH SIX GRAPHS

figure('Name','Microwave Transmission Line S-Parameter Analysis',...
       'NumberTitle','off',...
       'Position',[50 50 1400 900]);

%% GRAPH 1: COAXIAL TRANSMISSION LINE MODEL

subplot(3,2,1);

surf(X*1000,Y*1000,E,'EdgeColor','none');

xlabel('x (mm)');
ylabel('y (mm)');
zlabel('Electric Field (V/m)');
title('Graph 1: Coaxial Line Electric Field');
colorbar;
view(45,35);
grid on;
```

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axis tight;

%% GRAPH 2: PROPAGATION CONSTANTS

subplot(3,2,2);

yyaxis left;
plot(f/1e9,beta,'b','LineWidth',2);
ylabel('Phase Constant \beta (rad/m)');

yyaxis right;
plot(f/1e9,alpha,'r--','LineWidth',2);
ylabel('Attenuation Constant \alpha (Np/m)');

xlabel('Frequency (GHz)');
title('Graph 2: Propagation Constants');
grid on;

%% GRAPH 3: MAGNITUDE OF FOUR S-PARAMETERS

subplot(3,2,3);

plot(f/1e9,S11_dB,'b','LineWidth',1.8);
hold on;
plot(f/1e9,S21_dB,'r','LineWidth',1.8);
plot(f/1e9,S12_dB,'k--','LineWidth',1.8);
plot(f/1e9,S22_dB,'m-.','LineWidth',1.8);

xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Graph 3: S-Parameter Magnitudes');
legend('S11','S21','S12','S22','Location','best');
```

```
grid on;

%% GRAPH 4: PHASE OF S-PARAMETERS

subplot(3,2,4);

plot(f/1e9,unwrap(angle(S11))*180/pi,'b','LineWidth',1.5);
hold on;
plot(f/1e9,unwrap(angle(S21))*180/pi,'r','LineWidth',1.5);
plot(f/1e9,unwrap(angle(S12))*180/pi,'k--','LineWidth',1.5);
plot(f/1e9,unwrap(angle(S22))*180/pi,'m-.','LineWidth',1.5);

xlabel('Frequency (GHz)');
ylabel('Phase (degrees)');
title('Graph 4: S-Parameter Phase');
legend('S11','S21','S12','S22','Location','best');
grid on;

%% GRAPH 5: RETURN LOSS AND INSERTION LOSS

subplot(3,2,5);

plot(f/1e9,ReturnLoss,'b','LineWidth',2);
hold on;
plot(f/1e9,InsertionLoss,'r--','LineWidth',2);

xlabel('Frequency (GHz)');
ylabel('Loss (dB)');
title('Graph 5: Return and Insertion Loss');
legend('Return Loss','Insertion Loss','Location','best');
grid on;
```

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%% GRAPH 6: TRANSMISSION EFFICIENCY

subplot(3,2,6);

plot(f/1e9,Efficiency,'g','LineWidth',2);

xlabel('Frequency (GHz)');
ylabel('Transmission Efficiency (%)');
title('Graph 6: Transmission Efficiency');
grid on;
ylim([0 105]);

sgtitle('Numerical Simulation of Microwave Transmission Lines and Two-Port Networks');

fprintf('\nAll six graphs generated successfully.\n');
```

